

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2025

Programme: BE

Full Marks : 100

Course: Software Engineering Fundamentals (New)

Pass Marks : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are the attributes of good software? Explain prototype model with some advantage and disadvantage. 7
- b) Define metrics and Indicator with example. A Food Delivery Mobile Application is expected to require 45,000 LOC. The productivity rate is 180 LOC per person-month, and the labor rate is Rs. 18,000 per month. Find the effort and total project cost. 8

OR

Why is it necessary to estimate the project? Compute the function point value, effort, total cost of project for a project with following information domain characteristics:

No. of user input = 30

No. of user output = 20

No. of user inquiries = 8

No. of files = 7

No. of external interfaces = 6

Assume all complexity adjustment values are average and 14 algorithms have been counted. Productivity of software developers is 20FP/PM. And their salary is 1500/PM.

2. a) How do use cases and user stories help in understanding system requirements? Explain elements of analysis model. 8

OR

A bank plans to introduce an Online Banking System to provide services to its customers through the internet. Customers should be able to log in securely, view account balance, transfer funds to other accounts, and pay utility bills.

The system must validate every transaction with the bank's central database. The bank administrator manages customer accounts,

monitors *transactions*, and generates daily transaction reports. Show *generalization*, include and extend relationships if applicable.

Draw a Use Case Diagram showing interactions between actors and the system.

- b) Define Software Requirement Specification (SRS). Explain the essential components of a good SRS document. 7
3.
 - a) Discuss the importance of software architecture? Differentiate between transform mapping and Transaction mapping. 8
 - b) What is architecture design? What are the different types of architectural style? 7
4.
 - a) Define cyclomatic complexity. Using Basic path testing approach draw the flow graph and find the cyclomatic complexity for the following code. 8


```

large    = X [10];
for ( I = 1, I <= n - 1, i++)
{
  If ( X [i] > large)
    large    = X [i];
}
          
```
 - b) Write down the importance of software testing. Differentiate between black box testing and white box testing. 7
5.
 - a) Define formal technical review. What are the steps of FTR? Explain how do you conduct FTR. 8
 - b) What is Software Configuration Management (SCM)? Explain version control and change management with examples. 7
6.
 - a) Explain object design process. 8
 - b) Explain domain analysis process in object oriented development. 7
7. Write short notes on: (**Any two**) 2×5
 - a) Four P's of software
 - b) Types of Software Risks
 - c) Capability Maturity Model (CMM)